

EEG Data Processing and Analysis Pipeline

Laboratory EEG arm

1 Ingestion / Landing: collect recordings and experiment logs



EEG recording

32 channel EEG · markers



Experiment log

source clock events

Link files to participant

participant ↔ experiment ↔ XDF

LAND

2 Bronze: preserve and verify raw files

Raw XDF + logs

Verify file integrity

Link XDF to log

Audit raw markers

DERIVE

3 Silver: align event timing and clean EEG

Align event clocks

Create event timeline

Estimate missing times

Event timing
valid?

NO

YES

Convert XDF to MNE

Filter + rereference

Repair bad channels

Signal usable?

NO

YES

4 Gold: build analysis ready EEG datasets

Condition windows

Ad and no ad windows

Extract 4 s features

Validate datasets

5 Analysis: compare conditions and report findings

Condition contrasts

Ad response contrasts

Results and figures